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Literature Review Report: Intelligent Traffic Systems & Machine Learning Paradigms

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1. Intelligent Traffic Systems (ITS)

Traditional Urban Traffic Control (UTC) systems rely on deterministic formulas or fixed-time schedules that fail to adapt to real-time, dynamic variations in traffic demand. **Intelligent Traffic Systems (ITS)** seek to resolve these structural limitations by leveraging decentralized computational frameworks. This literature review evaluates eight modern peer-reviewed papers (2023–2026) across two major paradigms in adaptive traffic signal control (ATSC): **Swarm Intelligence (SI)** and **Reinforcement Learning (RL)**.

2. Swarm Intelligence (SI) in Traffic Optimization

Swarm Intelligence approaches model traffic control as a global optimization problem where a population of simple agents (e.g., particles, ants) iteratively searches a complex space to locate optimal mathematical parameters (such as signal cycle lengths, phase sequences, and green splits).

A. Core Methodology

Modern SI literature focuses heavily on hybridizing classical global search algorithms with local refinement mechanisms or time-series prediction models:

- **Chaotic Mapping Hybridization:** (Ji, 2026) introduces a Regional Coordinated Traffic Control structure that dynamically partitions road networks using **License Plate Recognition (LPR)** data. It pairs a **Particle Swarm Optimization (PSO)** algorithm enhanced by a logistic chaotic map (to preserve population diversity and escape local minima) with a local sequential quadratic programming (SQP) solver to track optimal phase coordination.
- **Bi-Objective Non-linear Delay Correction:** (Wang, 2026) targets oversaturated peak hours by altering the classic Webster delay equation to include a non-linear queue growth term. An **Improved Particle Swarm Optimization (IPSO)** algorithm with dynamic inertia weights ($w_{max}=0.9, w_{min}=0.4$) and local disturbance mechanisms is deployed to solve a bi-objective model balancing minimal vehicle delay and minimal queue growth rates.
- **Temporal Forecasting Integration:** (Almutoory, 2026) blends global metaheuristics with deep networks by designing a hybrid **PSO-LSTM (Long Short-Term Memory)** architecture. Instead of direct real-time actuation, the swarm optimizes the underlying spatial hyperparameters of the LSTM network to maximize short-term traffic volume predictions on major corridors (e.g., Al Khoud Road, Muscat).

- **Two-Stage Metaheuristic Alternatives:** (Lee et al., 2024) develop a simulation-based two-stage optimization algorithm coupling **Integer-Constrained Adam (ICA)** with **Tabu Search (TS)** to compute optimal signal phase sequences and durations for arbitrary intersections. It bypasses population-heavy metaheuristics by applying a strict local tabu memory structure to avoid redundant cycles.

B. Datasets & Simulation Environments

- **Real-World Trajectory Datasets:** Massive License Plate Recognition (LPR) trajectory feeds and real-world high-density intersection sensor streams (Ji, 2026; Wang, 2026).
- **Empirical Field Collections:** Local camera-based traffic density recordings and physical induction loop infrastructure data (Almutoory, 2026; Lee et al., 2024).
- **Simulation Engines:** Synchro, VISSIM, and the standard Simulation of Urban MObility (SUMO) platform are utilized to construct fitness evaluations for the swarm loops.

C. Advantages & Limitations

- **Advantages:** Excellent global exploratory capabilities; straightforward implementation without massive dataset pre-training; mathematical tractability when optimizing explicit objective functions (e.g., phase coordination rates or Webster delay extensions); fast early-stage convergence speeds.
- **Limitations:** High computational latency due to iterative simulation calls (population evaluations); high risk of premature convergence (species stagnation) in raw implementations; highly sensitive to initialization parameters; struggles to adapt smoothly to millisecond-by-millisecond structural adjustments without recalculating the swarm loop.

3. Reinforcement Learning for traffic optimization (RL)

Reinforcement learning shifts the paradigm from global static parameter tuning to continuous, sequential decision-making. Traffic signals are modeled as Markov Decision Processes (MDPs) where decentralized or centralized agents observe states (queues, speeds), execute actions (changing phase splits), and map responses via feedback signals (rewards).

A. Core Methodology

Modern RL-TSC architectures integrate advanced deep feature representation layers, multi-modal perception models, and multi-agent decentralized topologies:

- **Attention-Guided Joint Optimization:** (Dai, 2026) proposes a joint control model using a **Double Deep Q-Network (DDQN)** that optimizes both phase sequence and phase duration concurrently. To preserve systemic safety, the Webster method establishes base thresholds, while a **Squeeze-and-Excitation (SE)** attention mechanism focuses the neural network on vital macroscopic and microscopic state details (e.g., localized vehicle delays vs. queue trends).
- **Safety-Constrained Proximal Policy Optimization:** (Reyad, 2026) tackles corridor-level coordination using a decentralized Multi-Agent Deep Reinforcement

Learning (MADRL) scheme based on **Proximal Policy Optimization (PPO)**. Crucially, the reward function moves beyond simple vehicle delay metrics by incorporating real-time crash prediction risks derived from Extreme Value Theory (EVT).

- **Multi-Modal Vision-Language Intervention:** (Pang, 2025) introduces **ViRL-TSC**, an approach that integrates deep reinforcement learning with Vision-Language Models (VLMs) to handle long-tail, unmapped traffic edge cases. The architecture follows a specialized *RL $\rightarrow$ VLM* workflow: the primary DRL controller operates autonomously under normal flows, but a VLM step intervenes to interpret unusual environmental features via open-ended reasoning when complex edge cases occur.
- **Stochastic Noise Injection for Policy Exploration:** (Ortiz, 2026) formulates an Advantage **Actor-Critic (A2C)** network fortified with continuous **Noise Injection (NI)**. Operating under a completely decentralized layout where individual traffic lanes run independent learning processes, the injection of parameterized parameter-space noise forces the actor-critic policy to maintain exploration and prevent policy collapse during unpredictable demand drops.
- **Dynamic Structural Modeling:** (Dong et al., 2026) outline a **Dynamic Graph-Enhanced Multi-Agent Reinforcement Learning** architecture, modeling neighboring intersections as nodes on a dynamically re-weighting spatio-temporal graph to preserve system wide coordination.

B. Datasets & Simulation Environments

- **Simulation Platforms:** Dominantly anchored on the open-source **SUMO (Simulation of Urban MObility)** platform natively coupled with Traci (Traffic Control Interface) APIs.
- **Vision/3D Synced Contexts:** High-fidelity 3D traffic simulation environments linked with text/image logs to prompt multi-modal vision-language models (Pang, 2025).
- **Safety Metrics:** Real-time crash telemetry streams constructed via Extreme Value Theory (EVT) analytical platforms (Reyad, 2026).

C. Advantages & Limitations

- **Advantages:** Exceptional real-time adaptability to unexpected traffic variations; model-free execution that does not depend on rigid mathematical delay formulas; decentralized structures scale effectively across massive grid lines; handles multi-modal state inputs directly (visual logs, velocity matrices).
- **Limitations:** Suffers from long, unstable training trajectories and initial policy instability; weak performance out-of-the-box before convergence; prone to catastrophic forgetting if spatial patterns shift too rapidly; highly vulnerable to sparse or delayed reward structures; requires significant local computing assets to deploy deep neural network inference engines directly to field controllers.

4. Key Observations

1. Swarm Intelligence (SI) Systems

- Solves for global parameters over an explicit timeframe.
- Highly reliable out-of-the-box; mathematically predictable.
- Stalls out inside highly complex local minima.
- Cuts travel delays by ~20–22% (Ji, 2026).

2. Reinforcement Learning (RL) Systems

- Executes local continuous state-action changes instantly.
- Exceptionally flexible; fits multi-modal state representations well.
- High training sample inefficiency; initial chaotic exploration.
- Drastically reduces queue waiting backlogs once fully stabilized.

5. References

1. Swarm Intelligence Papers

- Ji (2026): *Regional Coordinated Traffic Signal Control Based on Improved Chaotic Particle Swarm Optimization:* https://doi.org/10.3390/math14081374
- Lee et al. (2024): *Two-stage algorithm for traffic signal optimization and web-service system development:* https://digital-library.theiet.org/doi/full/10.1049/itr2.12542
- Wang (2026): *A study on signal timing optimization at oversaturated urban intersections during peak periods:* https://doi.org/10.1117/12.3101865
- Almutoory (2026): *Practical Swarm Optimization and Long Short-Term Memory for Short-Term Traffic Prediction and Management:* https://www.su.edu.om/jeiti-journal/index.php/main/article/download/33/17

2. Reinforcement Learning Papers

- Dai (2026): *Double deep network-based traffic signal optimization method for isolated intersections:* https://www.maxapress.com/article/doi/10.48130/dts-0026-0004?viewType=HTML
- Ortiz (2026): *Adaptive Traffic Signal Control Using Deep Reinforcement Learning with Noise Injection:* https://www.preprints.org/manuscript/202602.0953
- Dong et al. (2026): *Dynamic graph-enhanced multiagent reinforcement learning for traffic signal control:* https://doi.org/10.1117/12.3101834
- Pang (2025): *ViRL-TSC: Enhancing Reinforcement Learning with Vision-Language Models for Context-Aware Traffic Signal Control:* https://openreview.net/pdf?id=36xNdrIUXa
- Reyad (2026): *Multi-agent deep reinforcement learning-based decentralized adaptive traffic signal control for corridor-level safety optimization:* https://doi.org/10.1139/cjce-2025-020